

Fluid Dynamics I

Complete Notes

Fluid Statics · Kinematics · Dynamics · Bernoulli · Viscosity

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Table of Contents

1. Introduction to Fluid Dynamics	2
2. Fluid Statics	2
3. Kinematics of Fluid Motion	3
4. Equation of Continuity	4
5. Euler’s and Bernoulli’s Equations	5
6. Applications of Bernoulli’s Equation	6
7. Viscosity and Viscous Flow	7
8. Irrotational Flow and Stream Function	8
9. Dimensional Analysis	9
Quick Reference Cheat Sheet	10

1. Introduction to Fluid Dynamics

Definition — Fluid

A **fluid** is a substance that continuously deforms under an applied shear stress, no matter how small. Liquids and gases are both fluids.

Types of Fluids

- **Ideal (Perfect) Fluid:** Incompressible, inviscid (no viscosity)
- **Real Fluid:** Has viscosity and compressibility
- **Newtonian Fluid:** Shear stress $\propto$ rate of strain (e.g. water, air)
- **Non-Newtonian Fluid:** Does not follow Newton's law of viscosity
- **Compressible:** Density changes with pressure (gases)
- **Incompressible:** Density constant (liquids)

Key Fluid Properties

Property	Formula	Unit (SI)
Density	$\rho = m/V$	kg/m ³
Specific Weight	$\gamma = \rho g$	N/m ³
Specific Gravity	$S = \rho_f / \rho_w$	Dimensionless
Dynamic Viscosity	$\mu = \tau / (du/dy)$	Pa·s
Kinematic Viscosity	$\nu = \mu / \rho$	m ² /s
Pressure	$P = F/A$	Pa (N/m ²)

2. Fluid Statics

Definition — Fluid Statics

Study of fluids at **rest**. No relative motion between fluid layers, so no shear stress acts.

Hydrostatic Pressure

Key Formula

$$P = P_0 + \rho gh$$

where P_0 = pressure at surface, h = depth below surface.

Pascal's Law

Theorem — Pascal's Law

Pressure applied to a confined fluid is transmitted equally in all directions throughout the fluid.

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

Archimedes' Principle

Theorem — Archimedes' Principle

A body wholly or partially submerged in a fluid experiences an upward buoyant force equal to the weight of fluid displaced:

$$F_B = \rho_f g V_{\text{displaced}}$$

- $F_B > W \Rightarrow$ body floats
- $F_B = W \Rightarrow$ body is neutrally buoyant
- $F_B < W \Rightarrow$ body sinks

3. Kinematics of Fluid Motion

Definition — Kinematics

Study of fluid motion without considering the forces causing it.

Types of Flow

- **Steady Flow:** Fluid properties at any point do not change with time $\left(\frac{\partial}{\partial t} = 0\right)$
- **Unsteady Flow:** Properties change with time
- **Uniform Flow:** Velocity same at every point at a given instant
- **Non-uniform Flow:** Velocity varies from point to point
- **Laminar Flow:** Fluid layers slide smoothly (low Reynolds number, $Re < 2000$)
- **Turbulent Flow:** Chaotic mixing of layers ($Re > 4000$)
- **Rotational Flow:** Fluid particles rotate about their own axis
- **Irrotational Flow:** No rotation; curl of velocity = 0

Reynolds Number

Key Formula

$$\text{Re} = \frac{\rho V D}{\mu} = \frac{V D}{\nu}$$

$\text{Re} < 2000$: Laminar $\text{Re} > 4000$: Turbulent $2000 < \text{Re} < 4000$: Transitional

Velocity and Acceleration

For a velocity field $\mathbf{V} = u \hat{i} + v \hat{j} + w \hat{k}$:

Key Formula

Material (Substantial) Derivative:

$$\frac{D\mathbf{V}}{Dt} = \frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V} \cdot \nabla) \mathbf{V}$$

$$a_x = \frac{Du}{Dt} = \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z}$$

Streamlines, Pathlines, Streaklines

- **Streamline:** Curve tangent to velocity vector at every point at a given instant
- **Pathline:** Actual path traced by a fluid particle over time
- **Streakline:** Locus of particles that have passed through a given point

In **steady flow**, streamlines = pathlines = streaklines.

Equation of a Streamline

Key Formula

$$\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}$$

4. Equation of Continuity

Definition — Continuity Equation

Based on conservation of mass: mass can neither be created nor destroyed.

General Form (3D, Compressible)**Key Formula**

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$$

Or in vector form: $\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0$

For Incompressible Flow ($\rho = \text{const}$)**Key Formula**

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad \text{i.e.} \quad \nabla \cdot \mathbf{V} = 0$$

Continuity for 1D Pipe Flow**Key Formula**

$$A_1 V_1 = A_2 V_2 = Q \quad (\text{volume flow rate, m}^3/\text{s})$$

Worked Example

Water flows through a pipe of diameter 0.2 m at 3 m/s. Find velocity where diameter narrows to 0.1 m.

$$A_1 V_1 = A_2 V_2 \implies \frac{\pi(0.2)^2}{4}(3) = \frac{\pi(0.1)^2}{4}V_2 \implies V_2 = 12 \text{ m/s}$$

5. Euler's and Bernoulli's Equations**Euler's Equation of Motion****Derivation — Euler's Equation**

For an ideal (inviscid) fluid along a streamline:

$$\frac{\partial V}{\partial t} + V \frac{\partial V}{\partial s} = -\frac{1}{\rho} \frac{\partial P}{\partial s} - g \frac{\partial z}{\partial s}$$

In vector form:

$$\rho \frac{D\mathbf{V}}{Dt} = -\nabla P + \rho \mathbf{g}$$

Bernoulli's Equation

Theorem — Bernoulli's Equation

For steady, incompressible, inviscid flow along a streamline:

$$P + \frac{1}{2}\rho V^2 + \rho g z = \text{constant}$$

Or between two points:

$$P_1 + \frac{1}{2}\rho V_1^2 + \rho g z_1 = P_2 + \frac{1}{2}\rho V_2^2 + \rho g z_2$$

Bernoulli Terms

Term	Expression	Meaning
Static Pressure	P	Actual fluid pressure
Dynamic Pressure	$\frac{1}{2}\rho V^2$	Pressure due to motion
Hydrostatic Pressure	$\rho g z$	Pressure due to elevation
Total (Stagnation)	$P + \frac{1}{2}\rho V^2 + \rho g z$	Constant along streamline

Assumptions for Bernoulli: Steady flow · Incompressible · Inviscid · Along a streamline
· No energy added or removed

Worked Example — Bernoulli

Water flows from a tank through a pipe. At point 1: $V_1 = 2$ m/s, $z_1 = 5$ m, $P_1 = 100$ kPa. At point 2: $z_2 = 0$, $V_2 = 4$ m/s. Find P_2 .

$$P_2 = P_1 + \rho g(z_1 - z_2) + \frac{1}{2}\rho(V_1^2 - V_2^2) = 100000 + 1000(9.81)(5) + \frac{1}{2}(1000)(4 - 16) = 143050 \text{ Pa}$$

6. Applications of Bernoulli's Equation

Venturi Meter

Used to measure flow rate in a pipe.

Key Formula

$$Q = A_2 \sqrt{\frac{2(P_1 - P_2)}{\rho(1 - (A_2/A_1)^2)}}$$

Torricelli's Theorem

Speed of fluid exiting a tank through a small hole at depth h :

Key Formula

$$V = \sqrt{2gh}$$

Pitot Tube

Measures fluid velocity using stagnation pressure:

Key Formula

$$V = \sqrt{\frac{2(P_0 - P)}{\rho}}$$

where P_0 = stagnation pressure, P = static pressure.

7. Viscosity and Viscous Flow**Definition — Viscosity**

Viscosity is the measure of a fluid's resistance to deformation under shear stress.

Newton's Law of Viscosity**Key Formula**

$$\tau = \mu \frac{du}{dy}$$

where τ = shear stress (Pa), μ = dynamic viscosity (Pa·s), $\frac{du}{dy}$ = velocity gradient.

Navier-Stokes Equations

For incompressible viscous flow:

Key Formula

$$\rho \frac{D\mathbf{V}}{Dt} = -\nabla P + \mu \nabla^2 \mathbf{V} + \rho \mathbf{g}$$

x -component:

$$\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = -\frac{\partial P}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

Hagen-Poiseuille Flow (Pipe Flow)

Steady laminar flow in a circular pipe of radius R :

Key Formula

$$u(r) = \frac{1}{4\mu} \left(-\frac{dP}{dx} \right) (R^2 - r^2)$$

$$Q = \frac{\pi R^4}{8\mu} \left(-\frac{dP}{dx} \right) \quad V_{\max} = \frac{R^2}{4\mu} \left(-\frac{dP}{dx} \right)$$

In Hagen-Poiseuille flow: $V_{\text{avg}} = \frac{V_{\max}}{2}$. The velocity profile is parabolic.

8. Irrotational Flow and Stream Function

Vorticity and Irrotational Flow

Key Formula

Vorticity: $\omega = \nabla \times \mathbf{V}$

For 2D flow: $\omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$

Irrotational: $\omega_z = 0 \implies \frac{\partial v}{\partial x} = \frac{\partial u}{\partial y}$

Stream Function ψ

Definition — Stream Function

A scalar function $\psi(x, y)$ such that:

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}$$

Automatically satisfies continuity. Streamlines are curves $\psi = \text{const.}$

Key Formula

Flow rate between two streamlines:

$$Q = \psi_2 - \psi_1$$

Velocity Potential ϕ

Definition — Velocity Potential

For irrotational flow, $\exists$ scalar ϕ such that:

$$u = \frac{\partial \phi}{\partial x}, \quad v = \frac{\partial \phi}{\partial y}, \quad \mathbf{V} = \nabla \phi$$

Equipotential lines are curves $\phi = \text{const}$ and are **orthogonal** to streamlines.

Laplace's Equation

For irrotational, incompressible flow:

Key Formula

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0 \quad \text{and} \quad \nabla^2 \psi = 0$$

9. Dimensional Analysis

Buckingham π Theorem

Theorem — Buckingham π Theorem

If a physical problem involves n variables and m fundamental dimensions (M, L, T), then it can be expressed in terms of $(n - m)$ dimensionless π groups.

Important Dimensionless Numbers

Number	Formula	Physical Meaning
Reynolds (Re)	$\rho V L / \mu$	Inertia / Viscous forces
Froude (Fr)	$V / \sqrt{g L}$	Inertia / Gravity forces
Euler (Eu)	$P / \rho V^2$	Pressure / Inertia forces
Mach (Ma)	V / c	Flow / Sound speed
Weber (We)	$\rho V^2 L / \sigma$	Inertia / Surface tension

Quick Reference Cheat Sheet

FLUID DYNAMICS I — MASTER FORMULA SHEET

All key equations and results for the exam

Core Equations

Hydrostatic P	$P = P_0 + \rho gh$
Buoyancy	$F_B = \rho_f g V$
Continuity	$A_1 V_1 = A_2 V_2$
Bernoulli	$P + \frac{1}{2} \rho V^2 + \rho g z = C$
Torricelli	$V = \sqrt{2gh}$
Reynolds	$Re = \rho V D / \mu$
Viscosity	$\tau = \mu du/dy$
H-P Flow	$Q = \pi R^4 (-dP/dx) / 8\mu$

Stream Function & Potential

Stream fn	$u = \partial\psi/\partial y, v = -\partial\psi/\partial x$
Vel. potential	$u = \partial\phi/\partial x, v = \partial\phi/\partial y$
Laplace	$\nabla^2\phi = 0, \nabla^2\psi = 0$
Irrotational	$\partial v/\partial x = \partial u/\partial y$
Vorticity	$\omega_z = \partial v/\partial x - \partial u/\partial y$
Mat. Deriv.	$D/Dt = \partial/\partial t + \mathbf{V} \cdot \nabla$
Continuity	$\nabla \cdot \mathbf{V} = 0$
Euler	$\rho D\mathbf{V}/Dt = -\nabla P + \rho \mathbf{g}$

Dimensionless Numbers

$$Re = \frac{\rho V L}{\mu} \quad Fr = \frac{V}{\sqrt{gL}} \quad Ma = \frac{V}{c} \quad We = \frac{\rho V^2 L}{\sigma} \quad Eu = \frac{P}{\rho V^2}$$

$Re < 2000$: Laminar
 $Re > 4000$: Turbulent
 $2000 < Re < 4000$: Transitional

Flow Types — Quick Recall

- Steady: $\partial/\partial t = 0$
- Uniform: $\partial/\partial s = 0$
- Incompressible: $\nabla \cdot \mathbf{V} = 0$
- Irrotational: $\nabla \times \mathbf{V} = 0$
- Laminar: $Re < 2000$
- Turbulent: $Re > 4000$

Bernoulli Applications

- Venturi meter: flow rate in pipe
- Pitot tube: fluid velocity
- Torricelli: tank exit velocity
- Orifice: discharge coefficient
- Aerofoil: lift generation
- Syphon: fluid transfer

For feedback or to request additional topics:

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